Problem Formulation: Active Material Screening via Multi-Armed Bandits

1. Motivation and Problem Context

Discovering advanced functional materials—such as solid-state battery electrolytes, photovoltaic thin films, and high-entropy alloys—requires time-consuming wet-lab synthesis and characterization. Because experimental bandwidth, precursor materials, and testing equipment are severely constrained, exhaustive combinatorial screening is computationally and financially intractable. Active screening models this pipeline as an active decision problem: an agent iteratively selects which candidate composition to synthesize next, balancing the exploration of uncharacterized chemical spaces against the exploitation of formulations known to yield high physical performance.

2. Formal Problem Formulation

We formalize the experimental search as a stationary stochastic Multi-Armed Bandit tuple (S, A, R, E) over a discrete trial budget t = 1, 2, ..., T.

Environment (E)

A candidate library of K = 20 discrete candidate compositions. Each candidate material $i \in \{1, ..., K\}$ possesses an unknown true scalar property score $\mu_i \in [0, 1]$ (e.g., catalytic yield or conductivity). Physical testing introduces zero-mean Gaussian observation noise $\varepsilon_t \sim N(0, \sigma^2)$ with known experimental measurement error $\sigma = 0.05$.

State Space (S)

The environment is memoryless across independent trials, rendering the Markovian environment state trivial ($S_t = \varnothing$). The agent's decision state is fully represented by its interaction history $H_t = \{(a_1, r_1), ..., (a_{t-1}, r_{t-1})\}$, summarized by two running statistics for each candidate i:

- **Allocation count:** $N_i(t) = \sum_{\tau=1}^{t-1} \mathbb{1}(a_\tau = i)$
- **Empirical mean return:** $\hat{Q}_i(t) = (1 / N_i(t)) \cdot \sum_{\tau=1}^{t-1} r_\tau \cdot \mathbb{1}(a_\tau = i)$

Action Space (A)

A discrete action set $A = \{1, 2, ..., K\}$ of cardinality $|A| = 20$, where selecting action $a_t = i$ commits laboratory resources to synthesize and characterize candidate material i.

Reward Function (R)

Executing action $a_t = i$ returns a stochastic scalar measurement:
$r_t = \mu_i + \varepsilon_t$, where $\varepsilon_t \sim N(0, \sigma^2)$

The agent's objective is to maximize cumulative experimental yield $E[\sum_{t=1}^{T} r_t]$, which corresponds directly to minimizing expected cumulative regret against the theoretical best candidate $\mu^* = \max_j \mu_j$:
$R_{regret}(T) = T \cdot \mu^* - \sum_{t=1}^{T} E[r_t] = \sum_{i=1}^{K} E[N_i(T)] \cdot (\mu^* - \mu_i)$

3. Implementation and Baseline Evaluation

- **Proposed Algorithm:** Upper Confidence Bound (UCB1), which resolves the exploration-exploitation tradeoff through optimism in the face of uncertainty:
    - $a_t = \text{argmax}_{i \in \{1, \ldots, K\}} [\hat{Q}_i(t) + c \cdot \sqrt{(\ln(t) / N_i(t))}]$
    - using exploration parameter $c = \sqrt{2}$. Each arm is pulled once during initialization ($t \leq K$).
- **Baseline Strategy:** A standard Uniform Random screening policy where $a_t \sim \text{Uniform}(\{1, \ldots, K\})$, reflecting unguided combinatorial trial-and-error experimentation.